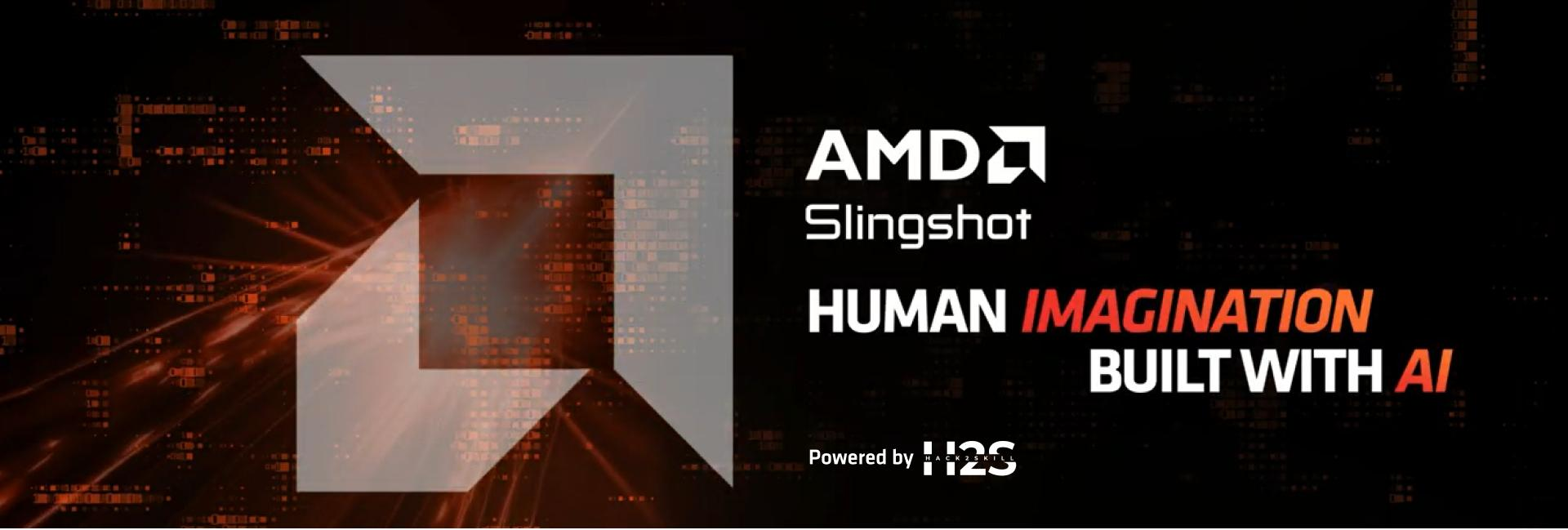


Team Details

- ➔ Team name - Trace_Minds
- ➔ Team leader name - Shreyas S
- ➔ Problem Statement - Kalpana: An AI conversational therapist that listens, understands, and connects you to someone who has been where you are

Brief about the idea

The Problem: Individuals experiencing emotional distress (like sudden job loss, grief, or severe isolation) face high barriers to immediate, stigma-free support. Existing AI chatbots often fail by offering unsolicited, robotic advice rather than genuine empathy, while professional help can be inaccessible in the moment of crisis.

Our Solution: Kalpana AI is a privacy-first, AI-driven mental health companion and intelligent peer-matchmaking platform.

How it Works (The Core Loop):

->Advice-Free Empathy: Kalpana provides a safe, grounded space for users to vent, using phase-based listening tailored purely to validate their feelings—never to "fix" them.

->Invisible Clinical Analysis: In the background, a secondary AI agent continuously maps the conversation, acting as a clinical psychologist to silently assess risk levels, self-harm indicators, and the underlying root cause of the distress.

->Semantic Peer Matchmaking: Once the system isolates a specific life event (e.g., "Bereavement") and a moderate risk threshold is met, Kalpana automatically connects the user with an anonymous peer who has survived the exact same experience.

->The Vision: To create an ecosystem where AI doesn't try to replace human connection, but rather acts as a compassionate bridge—providing immediate stabilization and then routing users to the profound healing of shared human experience.

Opportunities

How is Kalpana different from existing ideas?

->Most chatbots (e.g., Woebot, ChatGPT): Rush to "fix" the problem, offering generic, robotic advice that makes users feel unheard and pathologized.
->Most human therapy platforms (e.g., BetterHelp): Are prohibitively expensive, require scheduling, and lack immediate crisis intervention
->Kalpana's Difference: Kalpana acts strictly as a safe venting ground. It explicitly avoids giving unsolicited advice, instead offering grounded, phase-based empathy (Explore → Probe → Process) that matches human-level active listening, while running clinical diagnostics invisibly in the background.
-> National helplines like tele-MANAS are almost always full, requiring users to wait for a long time before connecting with someone

How will Kalpana solve the problem?

->Immediate Stabilization: Users receive zero-latency emotional support the exact moment they need it, lowering their immediate distress baseline (Risk Score mitigation).
->Automated Human Crux: Once stabilized, Kalpana solves the "loneliness epidemic" by using semantic vector search (Pinecone) to match the user's specific root cause (e.g., "Job Loss") with an anonymous peer who survived the exact same experience, closing the loop with shared human healing.

Unique Selling Proposition (USP) of the Proposed Solution

->Dual-Agent Architecture: A friendly "Listener Agent" interacts with the user, while a separated "Mapper Agent" acts as a silent Clinical Psychologist to dynamically extract intent, self-harm risk (1-10), and primary emotions.
->State-Locking Analysis: Kalpana intelligently isolates the single external life event causing distress (the "Root Cause") from the conversation and locks its state, preventing the AI from losing context over long sessions.
->Explainable AI (XAI): The Clinical Mapper provides deterministic summaries detailing exactly why a risk score was assigned, ensuring systemic transparency and safety.
->Absolute Privacy: Powered by local Large Language Models (LLMs) via Ollama, guaranteeing that highly sensitive emotional data never leaves the user's device or secure environment.

List of features offered by the solution

Core Functionality:

->Advice-Free Empathic Chat interface: A completely non-judgmental conversational space optimized for validating feelings rather than fixing them.

->Real-time Clinical Profiling: Background analysis that continuously calculates a 1-10 Risk Score and detects primary emotions on the fly.

->Automated Semantic Matchmaking: An intelligent routing system that finds and proposes anonymous real-world peers who have survived the exact same life-event root cause.

Advanced Technical Features:

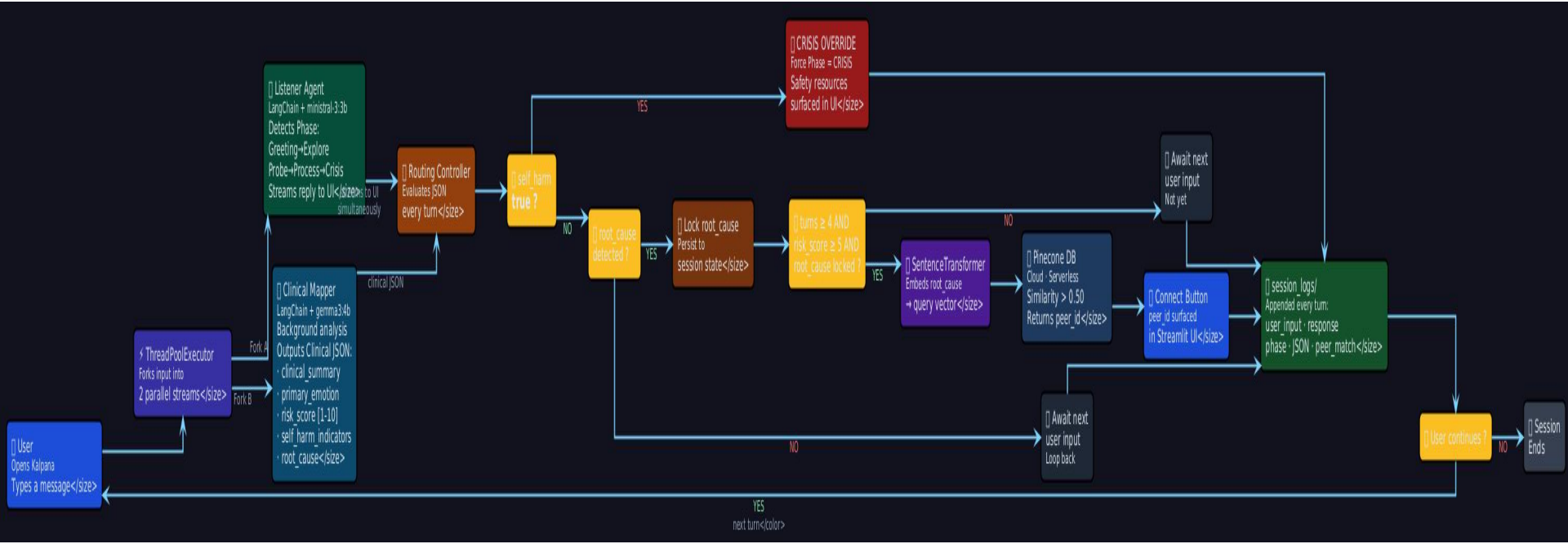
->Dual-Agent Processing: Decoupled LLM orchestration where ministral-3:3b handles fast, empathetic dialogue, and gemma3:4b executes heavy clinical mapping simultaneously.

->Explainable AI (XAI) Logging: The generation of deterministic clinical_summary records that explicitly justify the AI's diagnostic reasoning and risk assignment.

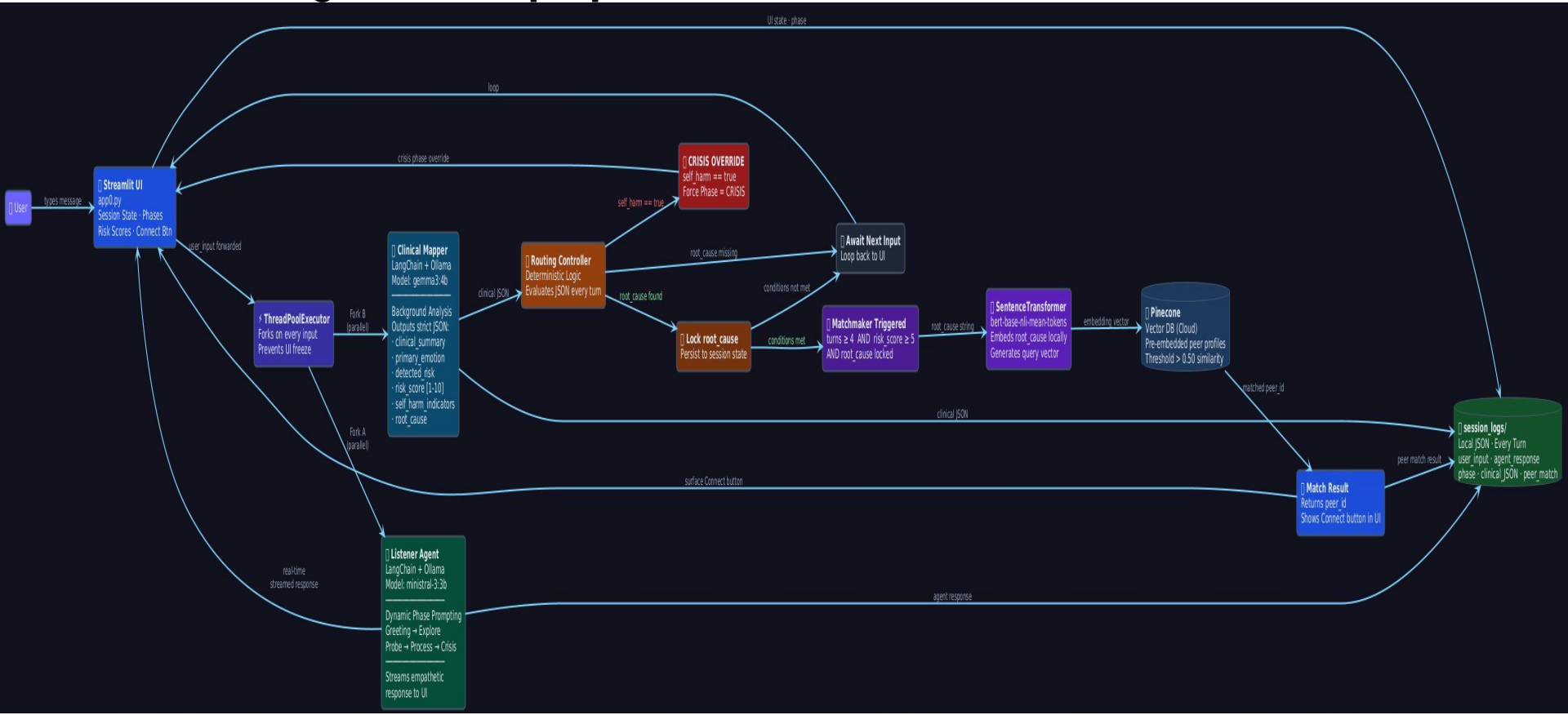
->Context State-Locking: The ability of the Mapper Agent to isolate and lock the underlying "Root Cause" of distress early in a session, preventing context loss over extended chats.

->Self-Harm & Safety Escalation: Immediate detection of explicit or implied critical risk indicators, triggering automated localized escalation (e.g., routing to the Tele-MANAS hotline).

Process flow diagram or U se-case diagram



Architecture diagram of the proposed solution



Technologies to be used in the solution

AI & Machine Learning Infrastructure:

->Local LLMs via Ollama: Ensuring absolute user privacy and zero data leakage.
-Ministral-3:3b (CPU-allocated): Drives the fast, empathetic Conversational Listener.
-Gemma3:4b (GPU-allocated): Drives the heavy-lifting Clinical Mapper for background analysis.

->LangChain: Orchestrating complex logic between the Listener and Mapper agents, managing conversation history, and handling phase transitions.

Vector Search & Matchmaking Engine:

->Pinecone DB: High-performance, scalable vector database used to store and retrieve peer profiles.
Hugging Face Transformers (sentence-transformers): Generating dense text embeddings of psychological root causes to semantically match distressed users with appropriate peers.

->Backend Architecture:
Python: The core runtime for AI agent logic.

->Frontend Architecutre:
Streamlit UI frontend UI interface

Usage of AMD Products/Solutions

Primary Development & Inference Hardware:

Processor: AMD Ryzen™ 7 7445HS
Integrated Graphics: AMD Radeon™ 740M Graphics
System Memory: 32.0 GB RAM
Architecture: 64-bit operating system, x64-based processor

How AMD Powers the Solution (Edge AI Execution):

Dual-Model Concurrency on AMD Hardware: Kalpana's unique dual-agent architecture relies entirely on the processing power of the AMD Ryzen 7 7445HS. It smoothly handles the simultaneous orchestration of two distinct Large Language Models natively on the device.

CPU Allocation for Real-Time Empathy: The lightweight Conversational Listener agent (ministral-3:3b) is explicitly allocated to run on the AMD CPU cores. This ensures zero-latency, high-speed text generation for immediate emotional validation without freezing the UI.

Compute/GPU Allocation for Diagnostics: Simultaneously, the heavier background Clinical Mapper agent (gemma3:4b), which performs complex semantic mapping and risk scoring, leverages the compute capabilities of the system, keeping the core operating environment fluid.

Absolute Privacy at the Edge: By using AMD hardware to run these models locally via Ollama, Kalpana guarantees that highly sensitive mental health data never leaves the user's device to hit a third-party cloud server.

Demo Video and Github Link

➔ GitHub Public Repository

https://github.com/sUhAs1011/AMD_SLINGSHOT

Video Link
https://drive.google.com/file/d/1ZrCBIwWpZz7AL8U1fO89qD3HNdeu6Wvr/view?usp=sharing

Future Plans & Technical Roadmap

While Kalpana AI currently operates as a fully functional V1 prototype demonstrating core dual-agent capabilities locally, our roadmap focuses on expanding its technical architecture and accessibility.

->Agentic Memory Management: enhancing the memory management so that we don't have to send conversation history to the model, while it can still give responses based on previous user prompts

->Safe Matchmaking & User History Database: Integrating a robust relational database to maintain encrypted user histories. This will enable complex matchmaking safety rails, explicitly preventing "co-rumination" (matching two users who might pull each other into a downward emotional spiral).

->Voice Integration & Multi-Lingual Support: Using Speech-to-Text (STT) and Text-to-Speech (TTS) models(Locally or through API) to enable natural voice note communication. This will be expanded to support native Indian languages, drastically lowering the barrier to entry for non-English speakers.

->LangGraph Orchestration & Pydantic Schema Grounding: Upgrading from linear LangChain prompts to a stateful LangGraph architecture for complex, multi-agent reasoning. We will enforce strict Pydantic schemas across all model outputs to guarantee 100% deterministic, accurate clinical profiling safely at scale.

->Frontend UI Modernization: Transitioning the core user interface from Streamlit to a fully custom React application. This will drastically improve state management, reduce latency, and allow for a more modular, responsive, and application-like user experience.

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S HACK2SKILL

Thank you!